Khanh N. Trinh

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EDUCATION

University of South Florida Tampa, FL
B.S. in Artificial Intelligence. Freshman. *May 2030*
Hanoi-Amsterdam High School for the Gifted Hanoi, Vietnam
High School Diploma (expected). GPA: Grade 9: 9.3 | Grade 10: 9.5 | Grade 11: 9.7. *June 2026*
Relevant Coursework: SAT: 1530 (Verbal 770, Math 760); SAT Superscore: 1570 (August 2025), IELTS: 7.5 Overall: Listening 8.0, Reading 9.0, Speaking 6.5, Writing 7.0 (October 2025)

PROJECTS

Diabetic Retinopathy Detection Using Machine Learning 2025 – May 2025
- Built a machine-learning pipeline to detect diabetic retinopathy from retinal fundus images using a publicly available dataset of labeled retinal images.
- Performed data preprocessing, model training, and evaluation using Python and computer-vision libraries, focusing on identifying visual patterns associated with diabetic retinopathy.
- Presented the system at the Advanced Innovation Global Competition in Singapore, receiving a Gold Medal.

Application of Machine Learning Algorithms in Prediction of Hotel Booking Cancellations September 2025 – September 2025
- Published research in IJSRM (International Journal of Scientific Research and Management), September 2025, evaluating five ML algorithms: Random Forest, k-Nearest Neighbors, Naive Bayes, SVM, and Decision Tree to predict hotel booking cancellations.
- Trained and tested models on 36,300 customer records (25,000 training, 11,300 testing) using 13 booking features including lead time, average price per room, meal plan type, and prior cancellation history.
- Random Forest achieved the highest classification accuracy at 91.0%, outperforming k-NN (82.5%), Naive Bayes (82.0%), SVM (80.0%), and Decision Tree (79.0%).

AI-Activated Robotic Arm 2025 – July 2025
- Developed the AI and programming components of a computer-vision-driven robotic arm, using an object-detection model to trigger physical arm movements based on detection results.
- Trained an object-detection model using the Ultralytics YOLO framework and wrote integration code connecting model predictions to robotic-arm behavior.
- Presented the system at the World Invention Creativity Olympic and Conference in Korea, receiving a Gold Medal.

ECO SMART RAIL TRANSIT: AI-Based Railway Safety Monitoring System 2025 – September 2025
- Built a computer-vision system for automated railway safety monitoring, training an object-detection model to identify potential hazards in railway environments.
- Managed the full ML pipeline: data preparation, model training using the Ultralytics YOLO ecosystem, and evaluation on held-out images; used cloud-based computing for model training.
- Presented the system at INOVA 2025 (49th International Invention Show) in Croatia, receiving a Gold Medal.

ACTIVITIES & LEADERSHIP

Society of Competitive Programmers (SCP), University of South Florida October 2026 – Present
Technical Lead

SKILLS

Programming Languages: Python, C++, JavaScript/TypeScript, HTML5, CSS, SQL
Machine Learning and AI: LangChain, NumPy, Pandas, Matplotlib, Seaborn, scikit-learn, PyTorch, TensorFlow, Ultralytics YOLO
Web/Mobile Technologies: Node.js, Vite.js, Express.js, React.js, FastAPI, React Native, Flask, Next.js, Tailwind CSS, SwiftUI
Tools: Git/GitHub
Technical: Machine Learning, Computer Vision, Object Detection, AI Algorithm Design, Hardware Prototyping, Arduino, Raspberry Pi, Sensor Modules
Languages: Vietnamese (Native), English (IELTS 7.5)

AWARDS & HONORS

- Gold Medal, Advanced Innovation Global Competition, Singapore (May 2025)
- Gold Medal, World Invention Creativity Olympic and Conference 2025, Korea (July 2025)
- Gold Medal, INOVA 2025 (49th International Invention Show), Croatia (September 2025)